



Qwen3-Max ▾



...

T

But note: each call to `smoothed_f` uses new random samples → high variance.

However, the prompt says:

""where $\hat{f}(x, y)$ is the Monte-Carlo estimate... using random samples""

So we compute:

- `f_x_plus = smoothed_f(x + delta, y, sigma, N)`
- `f_x = smoothed_f(x, y, sigma, N)`
- Same for y

✓ Implementation:

python



```
1 def finite_diff_grad_step(x, y, delta, N, eta, sigma):
2     f_x = smoothed_f(x, y, sigma, N)
3     f_x_plus = smoothed_f(x + delta, y, sigma, N)
4     f_y_plus = smoothed_f(x, y + delta, sigma, N)
5
6     df_dx = (f_x_plus - f_x) / delta
7     df_dy = (f_y_plus - f_x) / delta # Note: uses same f_x as baseline
8
9     new_x = x + eta * df_dx
10    new_y = y + eta * df_dy
11    return new_x, new_y
```

💡 Note: Some implementations use independent samples for each evaluation (as above). This is correct per instructions.



图像编辑

网页开发

图像生成

视频生成

预览模式

有什么我能帮您的吗？



深度思考

搜索



人工智能生成的内容可能不准确。